Query-Contrastive Grasp Fields for Resource-Constrained Language-Driven Grasp Detection

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Abstract

We study the quality objective of a compact language-driven grasp detector with frozen image and text encoders. Query-Contrastive Grasp Fields (QCGF) average foreground and background focal terms separately and rank target grasp regions above regions of other objects in the same training image. On scene-disjoint splits of a Grasp-Anything++ subset, QCGF reaches 47.4% harmonic-mean top-1 success over three seeds, 4.6 points above a matched MSE field. Matched class-balanced BCE and focal controls close most of that gap: the ranking term's gain over them is not significant with three seeds.

1. Introduction

A grasp can be geometrically valid and still target the wrong object. We train a small grasp-field decoder on Grasp-Anything++ [6], where the instruction identifies an object or part in the image. Grasp labels cover little of the quality map; we average grasp and background terms separately and rank the target above other objects' grasps. We test this objective against MSE, class-balanced BCE and focal controls with the architecture and budget fixed, and against the LGD authors' GR-ConvNet+CLIP retrained on our split. Code: https://github.com/hungtrab/qcgf-grasp.

2. Related Work

GR-ConvNet [2] predicts dense grasp quality, angle, and gripper width. LGD [6] conditions a diffusion model on language and adds a contrastive objective over diffusion states. QCGF instead compares annotated grasp regions from different objects in one image. Its decoder uses CLIP [5] text features, FiLM [4], and focal loss [3].

3. Method

Grasp field. Given image I and instruction t, we predict $g = (x, y, w, h, \theta)$, with θ periodic modulo π. Frozen ImageNet ResNet-18 [1] stages at strides 8, 16, and 32 encode a 224×224 image. A feature pyramid adds spatial coordinates; frozen CLIP ViT-B/32 text features condition it through FiLM. A decoder with 0.16M trainable parameters predicts quality logits z, $\cos 2\theta$, $\sin 2\theta$, and positive (w, h) on a 56×56 grid. At inference, the highest smoothed quality sets the grasp center; the other maps give its angle and size. No target annotations are supplied.

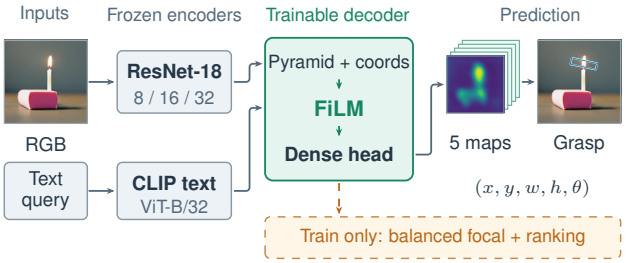


Figure 1. QCGF predicts an instruction-conditioned grasp field with frozen encoders. Target and distractor grasp supports supervise training only.

Quality objective. Let P be the union of central-third support regions of valid target grasps, and N its complement. With $p_u = \sigma(z_u)$, we average focal loss separately:

$$\mathcal{L}_q = -\frac{1}{|P|}\sum_{u\in P}(1-p_u)^2 \log p_u - \frac{1}{|N|}\sum_{u\in N} p_u^2 \log(1-p_u). \quad (1)$$

Let D contain grasp support from other objects in the same training scene, excluding overlap with P and excluding same-object parts. Define $s(S) = \log \sum_{u\in S} \exp z_u - \log |S|$. The log-mean-exp scores normalize for support size. The ranking term is

$$\mathcal{L}_r = \mathrm{softplus}(s(D) - s(P) + 0.5)\,. \quad (2)$$

We average $\mathcal{L}_r$ over examples with target and distractor support; it is zero if none qualify. The total loss is $\mathcal{L} = \mathcal{L}_q + \mathcal{L}_\theta + 5\mathcal{L}_{wh} + 0.2\mathcal{L}_r$. Here $\mathcal{L}_\theta$ is cosine distance of normalized doubled-angle vectors and $\mathcal{L}_{wh}$ is smooth-L_1 size error, both averaged on P. Overlaps use the nearest grasp.

Table 1. Top-1 success (%) on our subset, mean ± sample standard deviation over three seeds. All rows were trained by us on the same split; GR-ConvNet+CLIP uses the LGD author code.

Method	Seen	Unseen	H
GR-ConvNet+CLIP	15.1±5.0	12.0±6.5	13.2±6.0
+ trained encoder	14.3±8.2	11.6±11.6	12.4±10.4
MSE field	39.0±1.3	47.4±3.1	42.7±2.0
Balanced BCE	48.2±1.2	42.2±8.2	44.7±4.7
Balanced focal	45.1±1.8	46.4±3.5	45.7±2.5
QCGF (focal+rank)	50.6±1.8	44.8±4.1	47.4±1.6

Controls replace $\mathcal{L}_q$ with MSE or class-balanced BCE, or keep focal $\mathcal{L}_q$ and drop $\mathcal{L}_r$.

4. Experiments

Protocol. Of 1,000/120 candidate training/validation scenes (seed 2026), 713/84 have usable part annotations. The authors' object lists define seen/unseen test queries. We keep scenes disjoint and assign any shared test scene to unseen, excluding its seen-object queries. Train/validation/seen-test/ unseen-test contain 2502/280/1376/601 queries. Success requires IoU > 0.25 and periodic angle error $< 30°$ against the same target grasp. We report top-1 rates and their harmonic mean H.

Training. All fields share initialization and batch order per seed (7, 17, 27) and train for 30 epochs with AdamW (10^{-3}, weight decay 10^{-4}), a cosine schedule and batch 32. The checkpoint with the highest validation top-1 is kept; test data select neither models nor hyperparameters. GR-ConvNet+CLIP uses the author code (commit `501904d`) with its loss, augmentation, Adam and batch 8 on 224^2 RGB; it trains up to 50 epochs and stops after 10 without validation gain. Its released network detaches visual features before text fusion, so its encoder never trains; we also report a variant without that detach. The controls were added after the first MSE/QCGF test results had been seen; all fields in Table 1 were then retrained on one CPU backend. An earlier GPU run gave 42.8% (MSE) and 49.0% (QCGF) H.

Results and limits. Paired differences in H (QCGF minus control, 95% bootstrap over scene clusters and seeds) are 4.6 [1.3, 8.2] against MSE, 2.7 [−2.7, 9.1] against balanced BCE and 1.7 [−2.2, 5.8] against balanced focal. Most of the gain over MSE therefore comes from class balancing, which the BCE and focal controls also provide; ranking's added effect is positive on average but uncertain, and it is concentrated in one seed. The released GR-ConvNet+CLIP reaches 13.2% H, and training its encoder gives 12.4% (a lower bound: two seeds stopped early while seed 7 improved to epoch 46). Both learn vision from 2.5k queries; the gap to our ImageNet-feature fields is not evidence about the quality loss. Replacing the instruction with another object's in the same image lowers QCGF seen success from 51.5% to 22.5% on 589 queries, so predictions depend on the text. We do not rank against full-data LGD ($H = 45\%$ [6]) or measure physical grasp success.



Figure 2. Target grasps (green), MSE field (orange), QCGF (cyan), from the GPU run. The first baseline failure corrected by QCGF (left) and first QCGF failure (right), in query ID order.

5. Conclusion

Balancing grasp and background terms is the main source of QCGF's improvement over an MSE grasp field on this subset; the same-scene ranking term needs more seeds or data to be confirmed. Full-dataset and physical trials remain open.

References

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